

Random Variables, examples

CSU, Chico Math 314

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outline

Discrete Distributions

Continuous Distributions

- Probabilities in R via the normal distribution

- Normal Rule

- Percentiles

- Integrate to 1

References

Probability, Uniform Distribution

What's the probability of rolling one die greater than 3?

- ▶ What's the probability mass function of rolling dice?
- ▶ We want to calculate $P(X > 3)$.

Probability, Uniform Distribution

Let $X \sim U(1, 6)$. Find $P(X > 3)$.

$$P(X > 3) = \sum_{x>3} P(X = x) = 3/6.$$

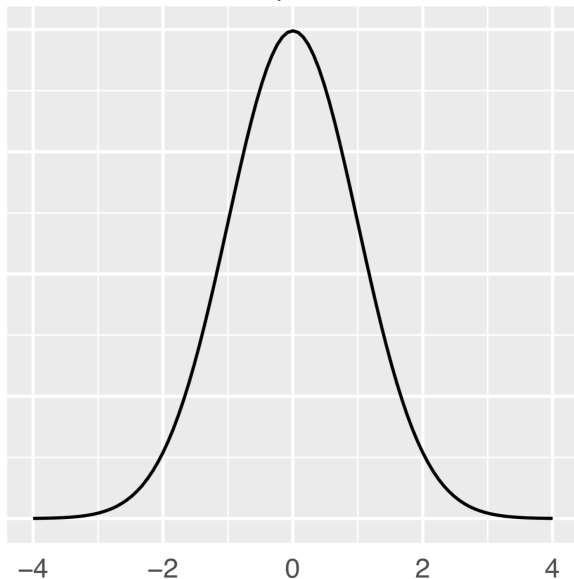
Discrete CDF

If the random variable X has the following cumulative distribution function, calculate the PMF.

x	1	2	3	4
$F(x)$	0.1	0.4	0.6	1.0

Normal Distribution

The **normal distribution** is ubiquitous in statistics.



Normal Distribution

The probability density function for the normal distribution is

$$f(x) = (2\sigma^2\pi)^{-1/2} \exp \left\{ \frac{-(x - \mu)^2}{2\sigma^2} \right\}.$$

Probability statements about the normal distribution take the form

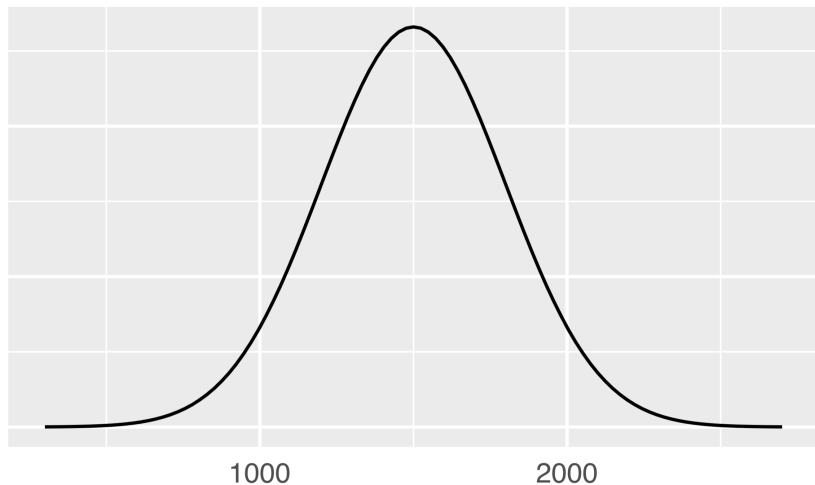
$$P(a < X < b) = \int_a^b f(x) dx.$$

Normal Distribution, example

A well known example of the normal distribution is SAT scores. Let $X \sim N(1500, 300^2)$ be the distribution of 2005-2016 SAT scores (scored out of 2400) with population mean $\mu = 1500$ and population variance $\sigma^2 = 300^2$.

Normal Distribution, example

SAT scores: $X \sim N(1500, 300^2)$.



Z-score

Z-score

The Z-score of an observation is the number of standard deviations the observation lies above or below the mean. We compute the Z-score for an observation x that follows a distribution with mean μ and standard deviation σ using

$$z = \frac{x - \mu}{\sigma}.$$

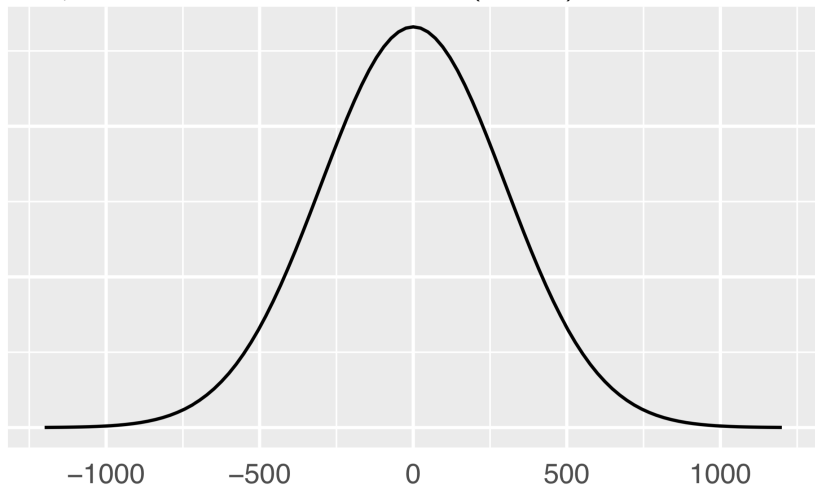
We'll often use the Z-score with data that are nearly normal, but this is a technique used for any distribution.

Z-score, notes

The Z-score simply shifts and scales the distribution so that z follows a $N(0, 1)$ distribution. Why?

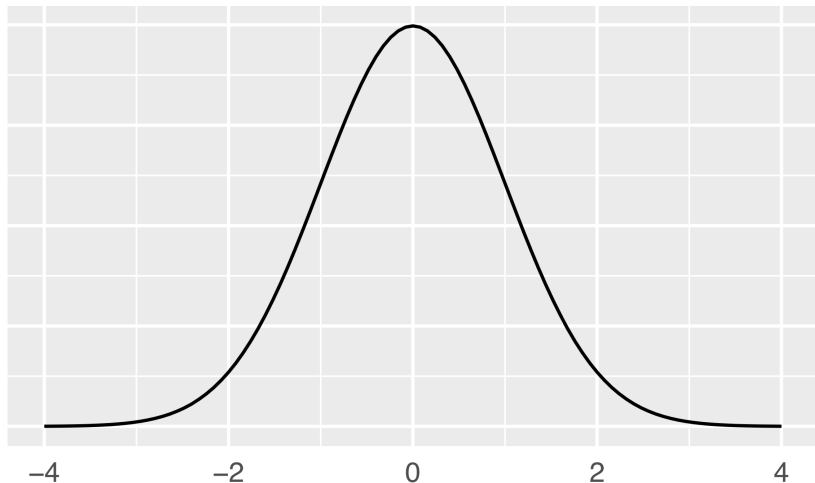
Z-score, shift

If SAT scores, X , are distributed $N(1500, 300^2)$, then $X - \mu = X - 1500$ is distributed as $N(0, 300^2)$.



Z-score, scale

If $(X - 1500) \sim N(0, 300^2)$, then $\frac{x-1500}{300} = z \sim N(0, 1)$.



Z-score, notes

The Z-score is a unitless number. Why?

Z-score, idea

By definition, if an observation is 1 standard deviation above its mean, the Z-score is 1. If an observation is 1.5 standard deviations below the mean, its Z-score is -1.5 .

Z-score, example

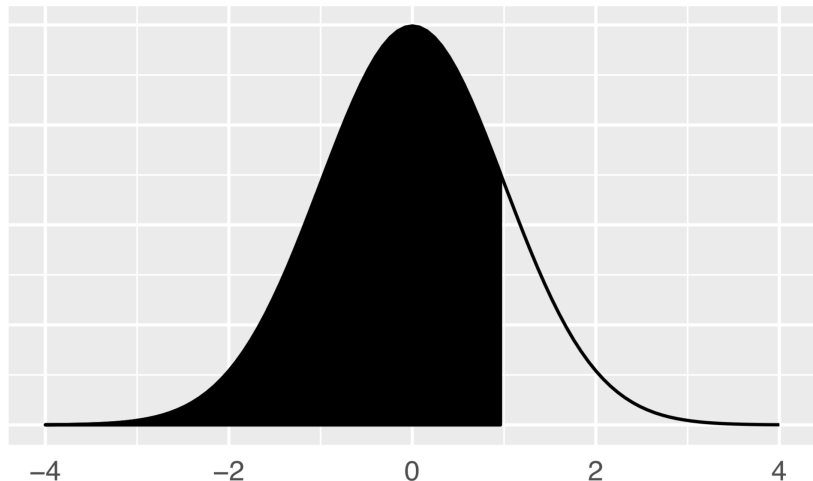
Suppose SAT scores follow a normal distribution, $X \sim N(1500, 300^2)$ and that Riley got 1800. What proportion of SAT takers did worse than Riley?

Steps

1. draw picture
2. Z-score: $z = \frac{1800-1500}{300} = 1$
3. $P(Z < 1)$?

Z-score, example

Picture of proportion of SAT takers that did worse than Riley:
 $P(Z < 1)$



Z-score, finding probability

Use R to find $P(Z < 1)$.

```
pnorm(1) # =  $P(Z < 1)$ 
```

```
## [1] 0.8413447
```

pnorm

The function `pnorm` in R calculates for us the area under the curve that's less than the value provided.

```
?pnorm
```

What is the stats name of this function?

Arithmetic with Densities

Since all the area under the normal curve is equal to one, we can easily manipulate the output of `pnorm` to find areas greater than some value, or in between two values.

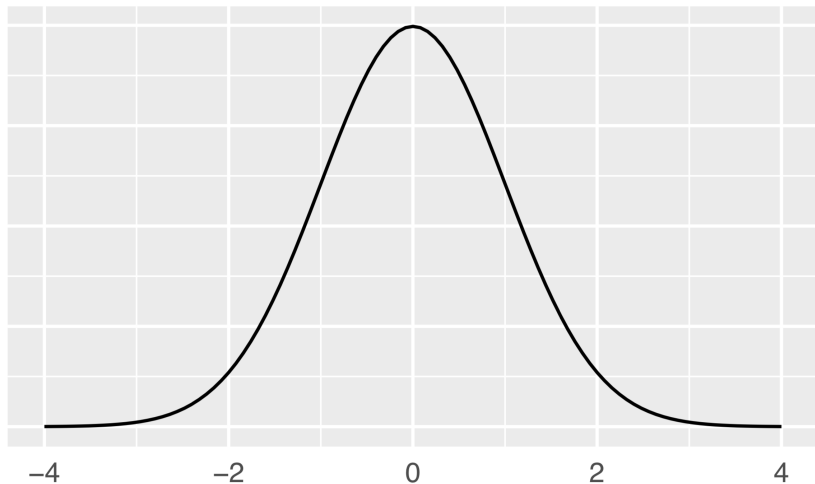
Z-score, example 2

April is a randomly selected SAT taker, and nothing is known about April's SAT aptitude. What is the probability April scored at least 1630 on the SATs?

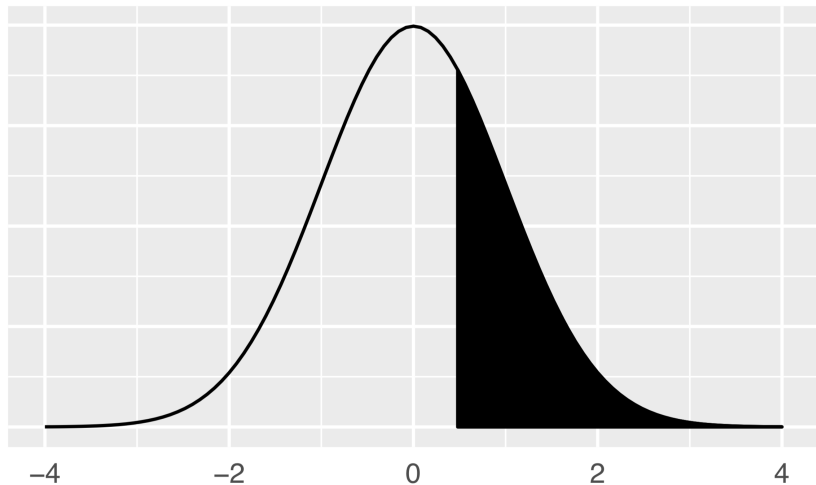
Steps

1. draw picture.
2. Z-score: $z =$
3. $P(Z > z)$?

Z-score, example 2



Z-score, example 2



Z-score, finding probability

We get $z = 0.43$. Use R to find $P(Z > 0.43)$.

```
1 - pnorm(0.43) # = P(Z > 0.43)
```

```
## [1] 0.3335978
```


Z-score, example 3

What's the probability that Ollie scored in between Riley and April? What would the picture for this be?

Z-score, finding probability

$P(0.43 < z < 1)$?

```
pnorm(1) - pnorm(0.43)
```

```
## [1] 0.1749426
```

This is R's version of

$$P(a < Y < b) = \int_a^b f_Y(y) dy = F(b) - F(a).$$

Z-score, another example

Because the Z – *score* is unitless, we can compare the likelihood of different events even if they are measured in different units.

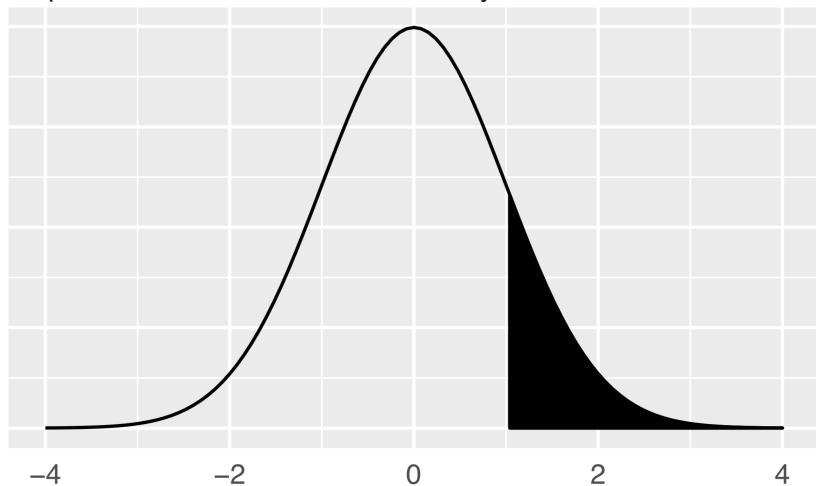
Z-score, another example

Suppose SAT scores are distributed $X \sim N(1500, 300^2)$ and ACT scores are distributed $Y \sim N(21, 5^2)$. If Riley scored 1800 on the SAT and Andy scored 24 on the ACT, who performed better?

1. draw pictures.
2. Z-scores: $z_A = 1$ and $z_T = \frac{24-21}{5} = 0.6$.
3. Want to find $P(Z > 1)$ and $P(Z > 0.6)$.

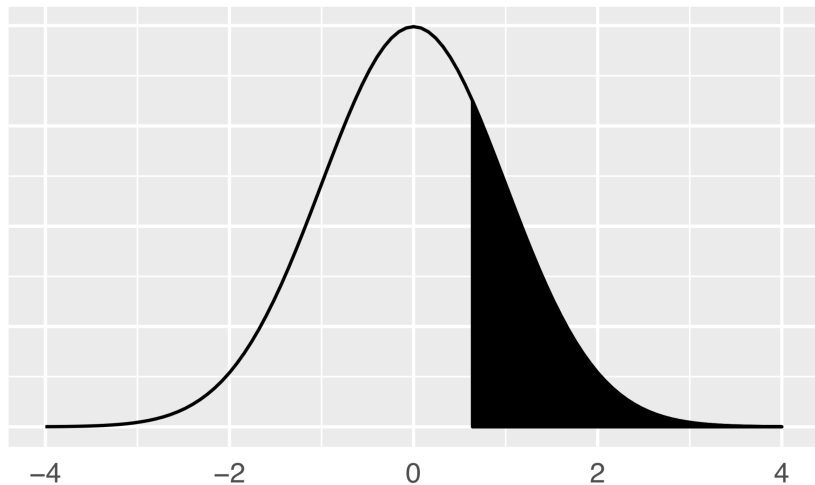
Z-score, example

Proportion who scored better than Riley



Z-score, example

Proportion who scored better than Andy



Z-score, finding probability

Use R to find $P(Z > 1)$ and $P(Z > 0.6)$.

```
1 - pnorm(1) # = P(Z>1), Riley
```

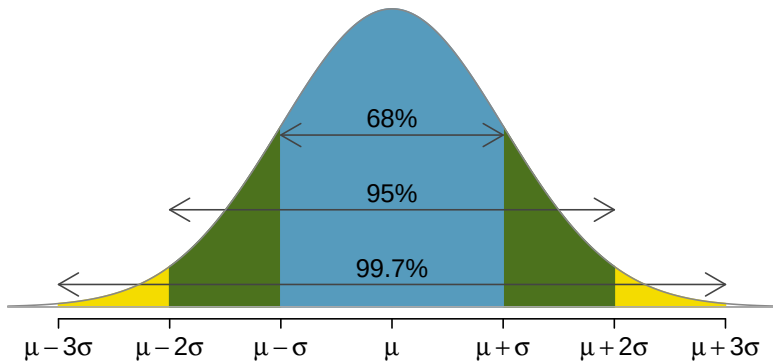
```
## [1] 0.1586553
```

```
1 - pnorm(0.6) # = P(Z>0.6), Andy
```

```
## [1] 0.2742531
```

Conclusion?

Normal Distribution, 68, 95, 99.7 Rule



Percentiles

Ollie earned a 1400 on the SAT. What is the percentile for this score?

Steps

1. draw picture.
2. Z-score: $z = (1400 - 1500)/300 = -0.33$.
3. $P(Z < -0.33)$?

Percentiles

The probability $P(Z < -0.33)$ can also be thought of as a percentile.

```
pnorm(-0.33)  
## [1] 0.3707
```

Ollie is at the 37th percentile.

Percentiles

We can essentially ask the previous questions backwards.

Percentiles, example

Gary's SAT score is at the 90th percentile. What score would this be?

Steps

1. Draw picture.
2. find Z-score.
3. solve for SAT score.

Percentiles, example

Find Z-score

```
qnorm(0.9) # finds Z-score z that puts 90% <
## [1] 1.281552
```

Percentiles, example

With $z = 1.28$, we can solve for Gary's SAT score.

$$z = 1.28 = \frac{x - 1500}{300}.$$

qnorm

The function `qnorm` in R calculates for us the Z-score z that puts the supplied probability less than z .

```
?qnorm
```

Distribution Functions in R

Notice that all random variables in R follow the same naming convention:

```
?pexp  
?pnorm  
?punif  
?pbinom # to get the Bernoulli use size=1
```

By this logic, there are equivalents for other such functions (`rexp()`, `qexp()`, etc.). Recall which ones refer to the PDF or CDF, and what questions apply them.

Exponential Distribution, integrates to 1

Recall, all distribution functions sum or integrate to one. Let's try one. Calculate

$$\int_0^{\infty} \lambda \exp(-\lambda x) dx.$$

references I

Adapted from notes by Dr. Roualdes.